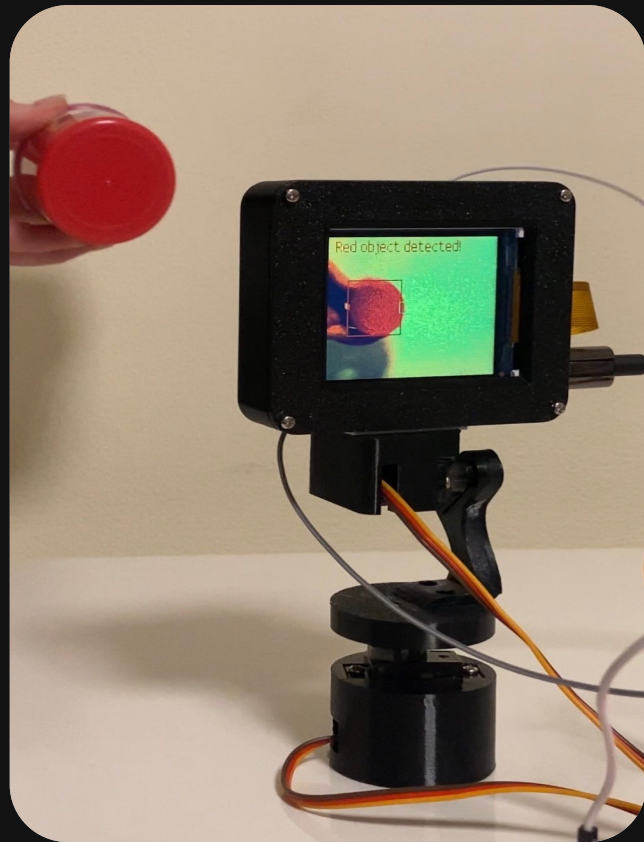


Edge Vision Pan-Tilt Tracker

real-time object tracking system using Maixduino, camera module, STM32 NUCLEO-F446RE, UART communication, and PWM servo control



Idea

Building a tracker that recognizes a red object in the frame and automatically rotates the camera to follow it along two axes.

Components

Component	Quantity
Maixduino	1
OV2640 Camera	1
LCD	1
STM32 NUCLEO-F446RE	1
MG90S Servo	2
1000 μ F, 10–16 V capacitor 1	1
AC/DC 5 V 2–3 A	1
DC female connector	1
Dupont wires	≥ 12

Architecture

The camera sends the image to Maixduino. Maixduino detects the object, determines its coordinates, and calculates the required Pan and Tilt angles. The angle values are transmitted via UART to the STM32 NUCLEO, which generates PWM signals for the two servos.

Maixduino + camera + display:

- detects the red object
- calculates the location of an object in the image
- sends the coordinates to the NUCLEO via UART

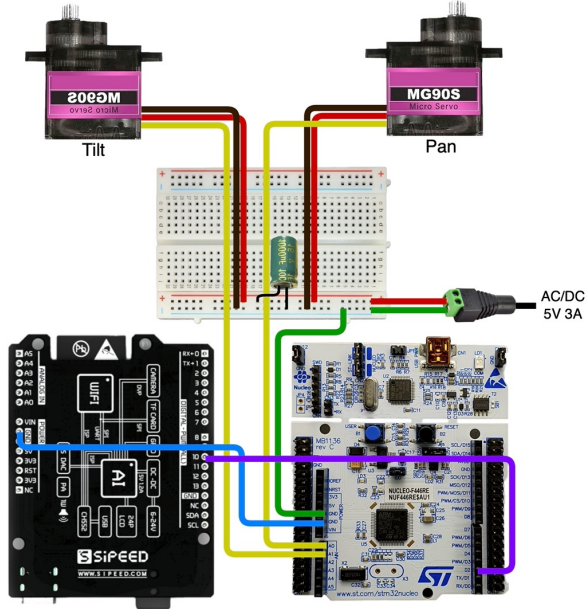
MG90S:

Pan servo – movement left-right, Tilt servo – movement up-down. Rotation to specified angles.

NUCLEO-F446RE:

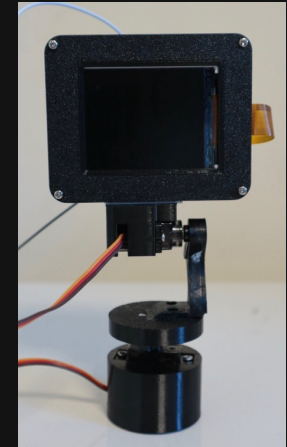
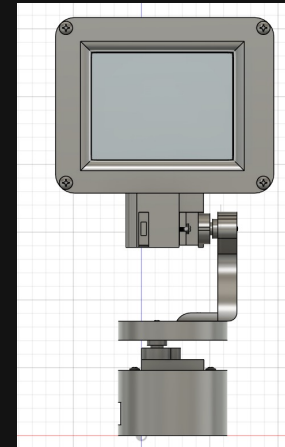
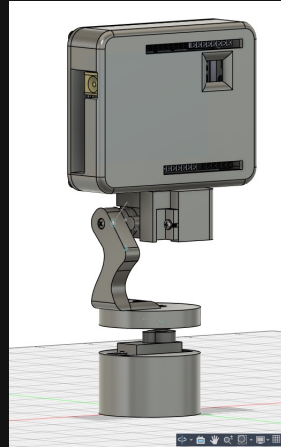
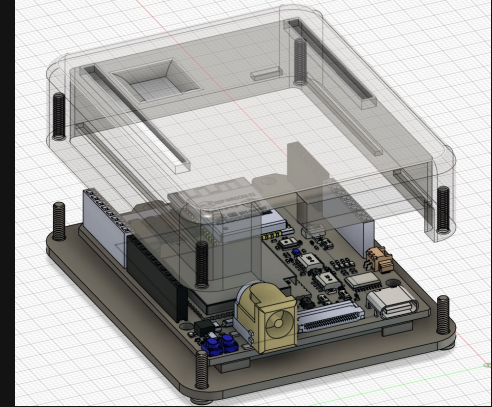
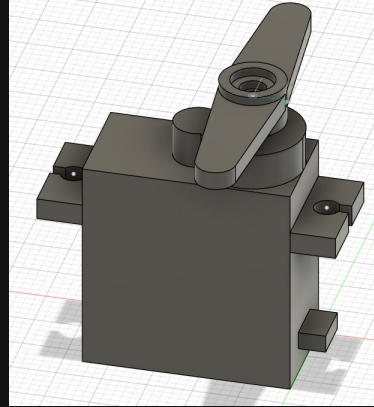
- receives the coordinates
- decides which pan-tilt axis to rotate
- outputs PWM to 2 MG90S servos

Connections



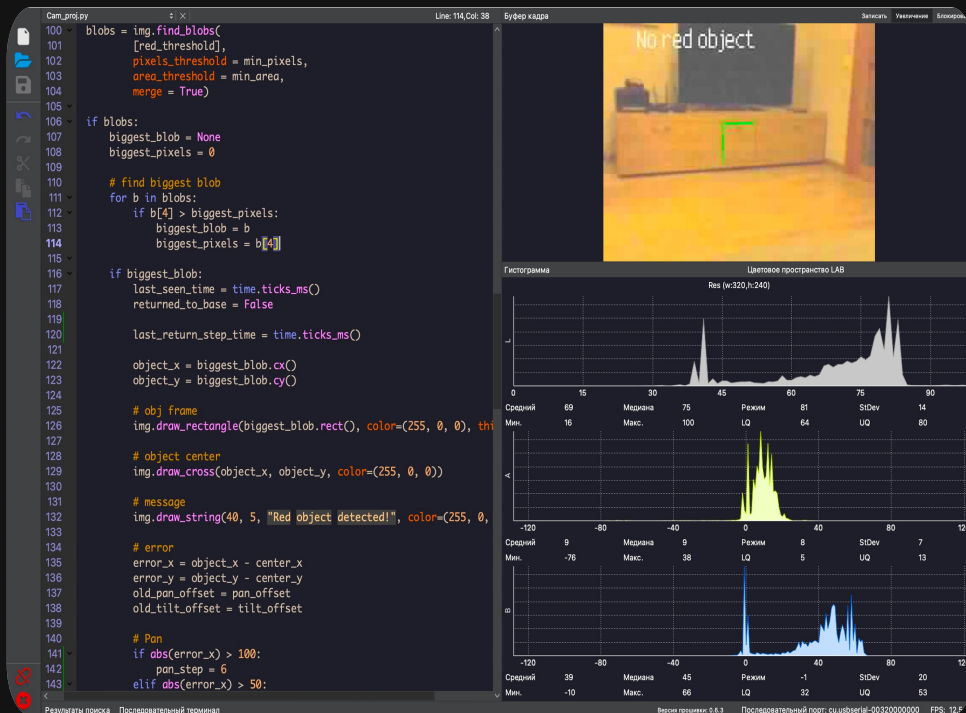
3D Modeling

Since none of the 3D designs from the internet were suitable for my servos and board, I designed and printed my own Pan-Tilt mechanism and Maixduino enclosure in Fusion 360. The Maixduino board with the camera and the display on the other side is mounted on top. Below are the servos that rotate. For easier mounting, M2-M2.5 screw holes were added throughout



Maixduino

Maixduino receives a 320×240 frame and finds the largest area that matches the specified LAB threshold for the red color. The center of the object and its deviation from the center of the frame (160, 120) are determined. Based on the error, pan_offset and tilt_offset are changed, after which the calculated angles are transmitted to the STM32.

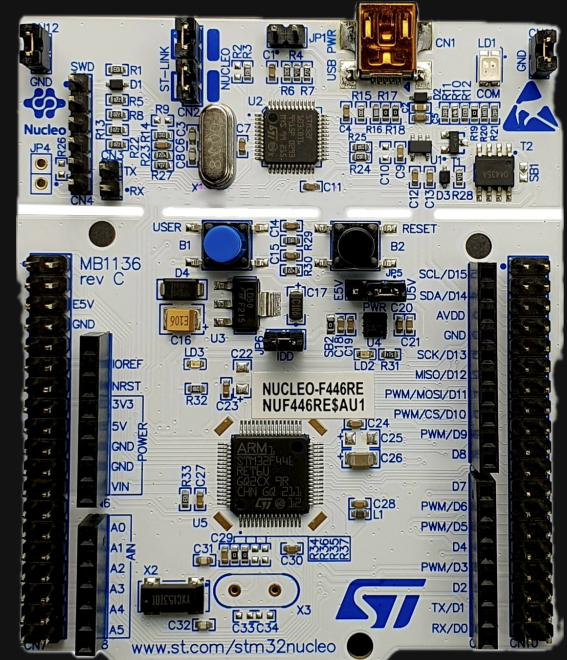


STM32 NUCLEO

The NUCLEO continuously waits for data via UART.

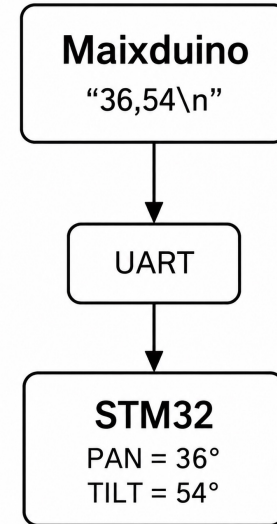
The command has the following format: "PAN,TILT\n".

STM32 parses the received values, checks the valid range, and updates the PWM for the Pan and Tilt servos.



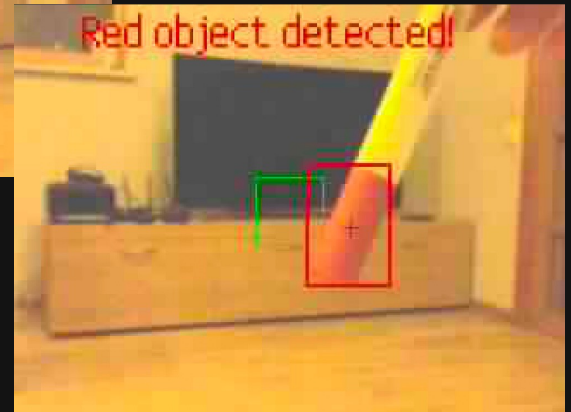
UART Communication

UART with a baud rate of 115200 is used for communication.
Maixduino acts as the transmitter, while STM32 acts as the receiver.
The already calculated angle values are transmitted.



Red Object Detection

The camera image is processed in RGB565 format, while color detection is performed using a LAB threshold. find_blobs() finds the areas of the required color, and the largest object is selected. The LCD displays the bounding box, the center of the object, and the message: "Red object detected".



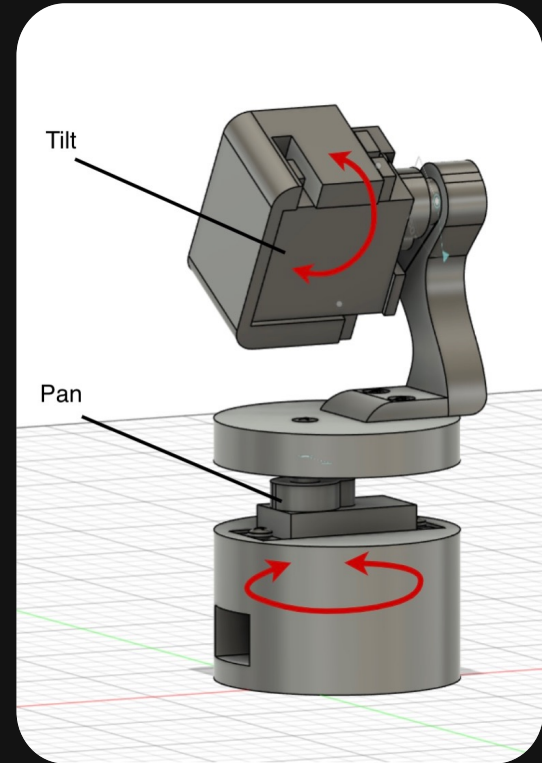
Servo Control

Pan – movement left-right, Tilt – up-down.

Rotation speed: the farther the object is from the center, the larger the rotation step.

Direction: controlled by the pan_dir and tilt_dir parameters (1 or -1).

- PWM frequency: 50 Hz
- Signal period: 20 ms
- Pulse duration: determines the servo angle
- Operating pulse range: 500–2500 μ s
- Base position: 90°
- Angle limit: 0–180°
- Maximum deviation from the base position: $\pm 90^\circ$



Result

<https://github.com/neura8/maixduino-stm32-object-tracker/blob/main/media/video/result.mov>